# Chapter II.3, Week 4

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3.10. *Fibres of a morphism.*

(a) If $f : X \to Y$ is a morphism and $y \in Y$ a point, show that $\mathrm{sp}(X_y)$ is homeomorphic to $f^{-1}(y)$ with the induced topology.

(b) Let $X = \operatorname{Spec} k[s,t]/(s - t^2)$, $Y = \operatorname{Spec} k[s]$, and $f : X \to Y$ the morphism defined by $s \mapsto s$. If $y \in Y$ is a point $a \in K$ with $a \neq 0$, show that the fibre $X_y$ consists of two points, with residue field $k$. If $y \in Y$ corresponds to $0 \in k$, show that the fibre $X_y$ is a nonreduced one-point scheme. If $\eta$ is the generic point of $Y$, show that $X_\eta$ is a one-point scheme, whose residue field is an extension of degree two of the residue field of $\eta$. (Assume $k$ algebraically closed.)

*Proof.* (a) We begin by considering the affine case, $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$, and $f^\sharp(Y) = \varphi : A \to B$. An identical method to that of Ex. 2.1 gives us the following lemma.

**Lemma 1.** *Let $A$ be a ring and $\phi : A \to S^{-1}A$ the inclusion into its localization. Then $\phi^* : \operatorname{Spec}(S^{-1}A) \to \operatorname{Spec} A$ is a homeomorphism onto its image in $\operatorname{Spec} A = X$. We denote this image $S^{-1}X$.*

The next lemma is easy, but worth proving.

**Lemma 2.** *Let $f : A \to B$ be a ring morphism, $X := \operatorname{Spec} A$, $Y := \operatorname{Spec} B$, and $S$ a multiplicative subset of $A$. Then denoting $S^{-1}B = f(S)^{-1}B \cong B_A \otimes_A S^{-1}A$ and identifying spectra as in the prior lemma,*

$$S^{-1}f^* : \operatorname{Spec}(S^{-1}B) \to \operatorname{Spec}(S^{-1}A)$$

*is the restrition of $f^*$ to $S^{-1}Y$, and $S^{-1}Y = f^{*-1}(S^{-1}X)$.*

*Proof.* We have the following commutative diagram.

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\phi_A"'] & B \arrow[d, "\phi_B"] \\ S^{-1}A \arrow[r, "S^{-1}f"'] & S^{-1}B \end{tikzcd}$$

Applying the contravariant functor Spec, we get a diagram of topological spaces.

$$\begin{tikzcd} X & Y \arrow[l, "f^*"'] \\ \operatorname{Spec}(S^{-1}A) \arrow[u, "\phi_A^*"] & \operatorname{Spec}(S^{-1}B) \arrow[l, "S^{-1}f^*"] \arrow[u, "\phi_B^*"'] \end{tikzcd}$$

Then

$$f^*(S^{-1}Y) = \phi_A^*(S^{-1}f^*(\operatorname{Spec}(S^{-1}B)) \subset S^{-1}X$$

allows us to restrict to the following diagram.

$$\begin{tikzcd} S^{-1}X & S^{-1}Y \arrow[l, "f^*|_{S^{-1}Y}"'] \\ \operatorname{Spec}(S^{-1}A) \arrow[u, "\phi_A^*"] & \operatorname{Spec}(S^{-1}B) \arrow[l, "S^{-1}f^*"] \arrow[u, "\phi_B^*"'] \end{tikzcd}$$

By the preceding lemma, $\phi_A^*$ and $\phi_B^*$ are heomeomorphisms onto their image, whence the result follows. $\square$

One last lemma precedes the proof of the desired statement.

**Lemma 3.** *Let $\mathfrak{a}$ be an ideal of $A$ and let $\mathfrak{b} = \mathfrak{a}B$ be its extension in $B$. Let $f : A/\mathfrak{a} \to B/\mathfrak{b}$ be the homomorphism induced by $f$. If $\operatorname{Spec}(A/\mathfrak{a})$ is identified with its canonical image $V(\mathfrak{a}) \subset X$, and $\operatorname{Spec}(B/\mathfrak{b})$ with $V(\mathfrak{b}) \subset Y$, show that $\overline{f}^* = f^*|_{V(\mathfrak{b})}$.*

*Proof.* We have the following commutative diagram.

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\pi_A"'] & B \arrow[d, "\pi_B"] \\ A/\mathfrak{a} \arrow[r, "\overline{f}"'] & B/\mathfrak{b} \end{tikzcd}$$

Once again applying Spec,

$$\begin{tikzcd} X & Y \arrow[l, "f^*"'] \\ \operatorname{Spec}(A/\mathfrak{a}) \arrow[u, "\pi_A^*"] & \operatorname{Spec}(B/\mathfrak{b}) \arrow[l, "\overline{f}^*"] \arrow[u, "\pi_B^*"'] \end{tikzcd}$$

and restricting as before gives us our last diagram.

$$\begin{tikzcd} V(\mathfrak{a}) & V(\mathfrak{b}) \arrow[l, "f^*|_{V(\mathfrak{b})}"'] \\ \operatorname{Spec}(A/\mathfrak{a}) \arrow[u, "\pi_A^*"] & \operatorname{Spec}(B/\mathfrak{b}) \arrow[l, "\overline{f}^*"] \arrow[u, "\pi_B^*"'] \end{tikzcd}$$

By Exercise 2.18, $\pi_A^*$ and $\pi_B^*$ are homeomorphisms onto their images $V(\mathfrak{a})$ and $V(\mathfrak{b})$, so that the result follows. $\square$

Finally, we set $X = \operatorname{Spec} B$, $Y = \operatorname{Spec} A$, and fix $y = \mathfrak{p} \in \operatorname{Spec} A$. Clearly, we have homeomorphisms

$$\begin{aligned} X_{\mathfrak{p}} &= \operatorname{Spec}(B \otimes_A k(\mathfrak{p})) \\ &\cong \operatorname{Spec}(B \otimes_A A_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}) \\ &\cong \operatorname{Spec}(B_{A_{\mathfrak{p}}}/\mathfrak{p}B_{A_{\mathfrak{p}}}) \\ &\cong V(\mathfrak{p}B_{A_{\mathfrak{p}}}), \end{aligned}$$

where the last equality follows from Exercise 2.18 as before. Then equalities

$$\begin{aligned}
&\cong V(\varphi_{\mathfrak{p}}(\mathfrak{m}_{\mathfrak{p}})B_{A_{\mathfrak{p}}}) \\
&\cong (\varphi_{\mathfrak{p}}^*)^{-1}(V(\mathfrak{m}_{\mathfrak{p}})) \\
&\cong f_{\mathfrak{p}}^{-1}(\mathfrak{m}_{\mathfrak{p}}) \\
&\cong f^{-1}(\mathfrak{p}).
\end{aligned}$$

follow from minimality of the extension of an ideal, maximality of $\mathfrak{m}_{\mathfrak{p}} \trianglelefteq A_{\mathfrak{p}}$, and the second lemma respectively.[1]

Fix affine open covers $\{U_i = \operatorname{Spec} A_i\}$ of $X$ and $\{V_j = \operatorname{Spec} B_j\}$ of $Y$. Let $g : \operatorname{Spec} k(y) \to Y$ be the natural map, giving $\operatorname{Spec} k(y)$ the structure of a $Y$-scheme. We have

$$k(y) = \mathcal{O}_{Y,y}/m_y\mathcal{O}_{Y,y} \cong (B_j)_y/y(B_j)_y := k_j(y)$$

for all $y \in Y_j$.

Then $X_y$ is the gluing of the spaces $f^{-1}(Y_j) \times_{Y_j} \operatorname{Spec} k_j(y)$, over all $Y_j \ni y$. We can write

$$X_i \cap f^{-1}(Y_j) = \bigcup_{\substack{g_{ijk} \in X_i \\ D_i(g_{ijk}) \subset f^{-1}(Y_j)}} D_i(g_{ijk})$$

as a union of affines. Defining $f_{ijk} := f|_{D_i(g_{ijk})} : D_i(g_{ijk}) \to Y_j$, the affine case gives us that

$$\begin{aligned}
X_y &= \bigcup_{\substack{i,j,\, g_{ijk} \in X_i \\ D_i(g_{ijk}) \subset f^{-1}(Y_j)}} D_i(g_{ijk}) \times_{Y_j} \operatorname{Spec} k_j(y) \\
&= \bigcup_{i,j,k} D_i(g_{ijk})_y \\
&= \bigcup_{i,j,k} f_{ijk}^{-1}(y) \\
&= f^{-1}(y) \cap \bigcup_{i,j,k} D_i(g_{ijk}) \\
&= f^{-1}(y).
\end{aligned}$$

(b) We have that $X_a = \operatorname{Spec} A_a$, where

$$\begin{aligned}
A_a &= \frac{k[s,t]}{(s-t^2)} \otimes_{k[s]} \frac{k[s]_{(s-a)}}{(s-a)k[s]_{(s-a)}} \\
&\cong k[t] \otimes_{k[t^2]} \frac{k[t^2]}{(a-t^2)} \\
&\cong \frac{k[t]}{(a-t^2)}.
\end{aligned}$$

[1]I originally wrote this out a while ago, so I hadn't written it up again, let alone in simplied form, when we had our meeting. For some reason I thought (b) was the interesting part lol.

Then $\operatorname{Spec} A_a \cong V(a - t^2) = \{(\sqrt{a} + t, \sqrt{a} - t)\} \subset \operatorname{Spec} k[t]$. If $a \neq 0$, then this is a two point set, with residue fields given by

$$k_a(\sqrt{a} - t) = \frac{k[t]}{(\sqrt{a} - t)} \cong k,$$
$$k_a(\sqrt{a} + t) = \frac{k[t]}{(\sqrt{a} + t)} \cong k.$$

If $a = 0$, then $\operatorname{Spec} A_0 = \{t\}$ is singleton, and $t \in A_0$ is a nilpotent.

Finally, we consider

$$\begin{aligned} A_\eta &\cong \frac{k[s,t]}{(s - t^2)} \otimes_{k[s]} \frac{k[s]_{(0)}}{(0)k[s]_{(0)}} \\ &\cong k[t] \otimes_{k[t^2]} k(t^2) \\ &\cong k(s)[\sqrt{s}]. \end{aligned}$$

Then $A_\eta$ is a PID, so its only ideals are of the form $(f(\sqrt{s}))$ for $f(\sqrt{s}) = f_1(s) + f_2(s)\sqrt{s}$. Then $f\overline{f}(\sqrt{s}) \in k(s)$ defined in the obvious manner is invertible whenever $f \neq 0$, so $(f(\sqrt{s})) = 0, A_\eta$ gives us that $\operatorname{Spec} A_\eta$ is singleton. Furthermore, $(A_\eta)_{(0)} \cong k(\sqrt{s})$, which is a degree two extension over $k(\eta) = k(s)$, as desired. □

3.12. *Closed subschemes of* $\operatorname{Proj} S$.

(a) Let $\varphi : S \to T$ be a surjective homomorphism of graded rings, preserving degrees. Show that the open set $U$ is equal to $\operatorname{Proj} T$, and the morphism $f : \operatorname{Proj} T \to \operatorname{Proj} S$ is a closed immersion.

(b) If $I \subset S$ is a homogeneous ideal, take $T = S/I$ and let $Y$ be the closed subscheme of $X = \operatorname{Proj} S$ defined as image of the closed immersion $\operatorname{Proj} S/I \to X$. Show that different homogeneous ideals can give rise to the same closed subscheme. For example, let $d_0$ be an integer and let $I' = \oplus_{d \geqslant 0} I_d$. Show that $I$ and $I'$ determine the same subscheme.

We will see later that every closed subscheme of $X$ comes from a homogeneous ideal $I$ of $S$ (at least in the case where $S$ is a polynomial ring over $S_0$).

*Proof.* (a) Denote $Y = \operatorname{Proj} S$ and $X = \operatorname{Proj} T$. We have $\varphi(S_+) = T_+$, so that $U = X$ by definition. Fix homogeneous $g \in S_+$. Surjectivity of $\varphi : S \to T$ implies $\varphi_g : S_g \to T_{\varphi(g)}$ is surjective. Then $\varphi$ degree preserving gives us that $(S_g)_d = \varphi_g^{-1}((T_g)_d)$ for all $d \geqslant 0$, whence $\varphi_g : S_{(g)} \to T_{(\varphi(g))}$ is surjective. It follows that $f^\sharp : \mathcal{O}_X \to f_* \mathcal{O}_Y$ is surjective.

Surjectivity of $\varphi$ gives us $\varphi(\varphi^{-1}(\mathfrak{b}))$ for all $\mathfrak{b} \trianglelefteq T$, whence

$$\mathfrak{p} \supset \mathfrak{b} \iff \varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\mathfrak{b}).$$

Then

$$f(V(\mathfrak{b})) = V(\varphi^{-1}(\mathfrak{b})) \cap f(X)$$

shows that, in particular,

$$f(X) = V(\ker \varphi) \cap f(X) \subset V(\ker \varphi).$$

Take $\mathfrak{p} \in V(\ker \varphi)$, and $a \in \varphi^{-1}(\varphi(\mathfrak{p}))$. Then we have $a' \in \mathfrak{p}$ with $\varphi(a) = \varphi(a')$, whence $a - a' \in \ker \varphi \subset \mathfrak{p}$ implies $a \in \mathfrak{p}$. Thus, $\mathfrak{p} = \varphi^{-1}(\varphi(\mathfrak{p})) \in f(X)$ gives us $f(X) = V(\ker \varphi)$.

We conclude that

$$f(V(\mathfrak{b})) = V(\ker\varphi + \varphi^{-1}(\mathfrak{b}))$$

is closed in $\operatorname{Proj} S$, so that $f$ is a homeomorphism onto the closed set $f(X) = V(\ker\varphi)$.

(b) Let $\pi : S \to S/I$ and $\pi' : S \to S/I'$ be the natural projections. Since $I' \subset I = \ker\pi$, we can factor $\pi = \varphi\pi'$, where $\varphi : S/I' \to S/I$ is the identity $\varphi_d : S_d/I'_d = S_d/I_d \to S_d/I_d$ for $d \geqslant d_0$. Then Ex. 2.14(b) gives us that $\varphi^* : \operatorname{Proj}(S/I) \to \operatorname{Proj}(S/I')$ is an isomorphism of sheaves, so $(\pi')^*(\operatorname{Proj}(S/I')) = \varphi^*\pi^*(\operatorname{Proj}(S/I)) \cong \pi^*(\operatorname{Proj}(S/I))$ give the same closed subscheme structure. □

3.13. *Properties of Morphisms of Finite Type*

(a) A closed immersion is a morphism of finite type.
(b) A quasi-compact open immersion is of finite type.
(c) A composition of two morphisms of finite type is of finite type.
(d) Morphisms of finite type are stable under base extension.
(e) If $X$ and $Y$ are schemes of finite type over $S$, then $X \times_S Y$ is of finite type over $S$.
(f) If $X \xrightarrow{f} Y \xrightarrow{g} Z$ are two morphisms, and if $f$ is quasi-compact, and $g \circ f$ is of finite type, then $f$ is of finite type.
(g) If $f : X \to Y$ is a morphism of finite type, and if $Y$ is noetherian, then $X$ is noetherian.

*Proof.* (a) Let $f : X \to Y$ be a closed immersion, $\{U_i = \operatorname{Spec} A_i\}_{i\in I}$ an affine open cover of $X$, and fix $V = \operatorname{Spec} B \subset Y$ open affine. Then

$$\begin{aligned} f^{-1}(V) &= \bigcup_{i\in I} f^{-1}(V) \cap U_i \\ &= \bigcup_{\substack{i\in I,\ g_{ij}\in A_i \\ D_i(g_{ij})\subset f^{-1}(V)}} D_i(g_{ij}) \end{aligned}$$

is a union of open affines $D_i(g_{ij}) = \operatorname{Spec}(A_i)_{g_{ij}}$.

We have that $f^\sharp : \mathcal{O}_Y \to f_*\mathcal{O}_X$ is surjective. That is, $f^\sharp_{\mathfrak{q}} : B_{\mathfrak{q}} \to A_{B_{\mathfrak{q}}}$ is surjective for each prime ideal $\mathfrak{p}$ of $B$. Then

$$\begin{aligned} f(D_i(g_{ij})) &= \{f^\sharp(V)^{-1}(\mathfrak{p}) \in \operatorname{Spec} B : \mathfrak{p} \trianglelefteq A_i,\ \mathfrak{p} \not\ni g_{ij}\} \\ &= \{\} \end{aligned}$$

(d) Let $f : X \to Y$ be a scheme of finite type, and $Y \to Y'$ a map of schemes. □